MILITARY SPECIFICATION

# DIETHYLENE GLYCOL DINITRATE (DEGDN)
# (FOR USE IN AMMUNITION)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. - This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of one type of Diethylene Glycol Dinitrate (DEGDN) for use in the formulation of propellants. (See 6.1.)

## 2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. - Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

THIS DOCUMENT CONTAINS PROPRIETARY DATA AND WAS FURNISHED TO THE UNITED STATES GOVERNMENT PURSUANT TO CONTRACT NUMBER DAAK10-79-C-0094 BETWEEN RHEINMETALL GmbH AND THE DEPARTMENT OF THE ARMY DATED 22 FEB. 1979 AND THE RECEIVER IS SUBJECT TO THE PROVISIONS THEREOF AND AGREES TO THE OBLIGATIONS OF SUCH CONTRACT.

BENEFICIAL COMMENTS (RECOMMENDATIONS, ADDITIONS, DELETIONS) AND ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT SHOULD BE ADDRESSED TO: COMMANDER, US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND, ATTN. DRDAR-QA, DOVER, NEW JERSEY 07801 BY USING THE SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARING AT THE END OF THIS DOCUMENT OR BY LETTER.

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SPECIFICATIONS

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|---|---|
| MIL-D-23296 | Diethylene Glycol |
| MIL-A-48078 | Ammunition, Standard Quality Assurance Provisions, General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking For Shipment |
| MIL-STD-286 | Propellant, Solid; Sampling, Inspection and Testing |
| MIL-STD-1218 | ACS Chemicals |
| MIL-STD-1235 | Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

(Copies of specifications, standards, drawings, and publications listed above required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document(s) form a part of this specification to the extent specified herein. The issue of the documents



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which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | |
|---|---|
| ASTM E168 | General Techniques of Infrared Quantitative Analysis |
| ASTM E300 | Sampling Industrial Chemicals |
| ASTM D2905 | Statement on Number of Specimens Required to Determine the Average Quality of a Textile Material |

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS

| | |
|---|---|
| Title 49 | Transportation, Parts 100-199<br>Special Permit No. 5704 |

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402. Orders for the above publications should cite, "49 CFR 100-199 (latest revision)").

2.3 Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Manufacture. - Diethylene Glycol conforming with MIL-D-23296 shall be used in the manufacture of DEGDN.



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3.2 Material. - The explosive shall be DEGDN conforming with the following requirements.

3.2.1 Infrared. - An infrared spectrogram shall be obtained on the material for information purposes only in accordance with the method specified in 4.5.1. (See 6.6).

3.2.2 Nitrogen. - The nitrogen content shall be 14.1 percent minimum to 14.53 percent maximum when determined in accordance with 4.5.2.

3.2.3 Moisture. - The moisture content shall be 0.75 percent maximum when determined in accordance with 4.5.3.

3.2.4 Alkalinity. - The alkalinity content, calculated as sodium carbonate ($Na_2CO_3$), shall not exceed .004 percent when determined in accordance with 4.5.4.

3.2.5 Acidity. - The acidity content, calculated as sulfuric acid ($H_2SO_4$), shall not exceed .01 percent when determined in accordance with 4.5.4.

3.2.6 Stability. - The DEGDN shall not change the color of standard potassium iodide starch paper in less than ten (10) minutes at 82.2°C when determined in accordance with 4.5.5.

3.3 First article sample. - This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. - The best commercial practices shall be used in the formulation of the product furnished under this specification. Procedures and controls shall be employed which preclude conditions that may adversely affect the safety or reliability of explosives (i.e. contaminated vessels, impurities, dirt, chips, etc).



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - The inspection requirements specified herein are classified as follows.

a) First article inspection
b) Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of approximately 100 grams, or as otherwise specified by the Contracting Officer of DEGDN obtained by sampling as described in 4.4.3.1. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures, equipment and materials which will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. - The sample will be subjected by the Government (see Table 1) to any or all of the examinations or tests specified in 4.5.



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Table 1. First Article Tests

| Test | Requirement | Test Method |
|---|---|---|
| Infrared (info. only) | 3.2.1 | 4.5.1 |
| Nitrogen Content | 3.2.2 | 4.5.2 |
| Moisture Content | 3.2.3 | 4.5.3 |
| Alkalinity | 3.2.4 | 4.5.4 |
| Acidity | 3.2.5 | 4.5.5 |
| Stability | 3.2.6 | 4.5.6 |
| Workmanship | 3.4 | Visual |



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4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Lot formation. - A lot shall consist of one or more batches of DEGDN produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of DEGDN that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous (see MIL-A-48078).

4.4.2 Examination. - The examinations listed in Table 2 shall be performed on inspection lots as defined in 4.4.1. If the sample fails to comply with any of the requirements, the batch of the lot shall be rejected.

4.4.3 Testing. - PRECAUTION: This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the DEGDN. (See 6.3.)

4.4.3.1 Sampling. - The DEGDN shall be sampled in accordance with paragraph 7 of ASTM-E300 using a rubber dipper. Two representative samples of approximately 60 ml shall be removed from each batch of the lot. The DEGDN shall be transferred to a rubber bottle fitted with a stopper (see 6.3) and the sample shall be properly identified. Each sample shall be tested independently in accordance with the applicable methods given in 4.5. If either sample fails to meet any of the test requirements, the batch of the lot shall be rejected. MIL-STD-105 shall apply for the defect classification given in Table 2. Sampling for infrared spectra shall be in accordance with 6.6.

4.4.3.2 Cleaning of equipment. - After disposal of the DEGDN from the rubber sample bottle, the bottle shall be cleaned by thoroughly rinsing with acetone (see 6.3).



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4.4.3.3 Inspection equipment. - For the performance of all tests and examinations specified, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.4.3.4 Tests.

Table 2

Lot Acceptance Tests

| Test/Examination | Requirement | Defect Classification |
|---|---|---|
| Nitrogen | 3.2.2 | Major |
| Moisture | 3.2.3 | Major |
| Alkalinity | 3.2.4 | Major |
| Acidity | 3.2.5 | Major |
| Stability | 3.2.6 | Critical |
| Workmanship | 3.4 | Minor |

4.5 Test methods and procedures. - All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218.

4.5.1 Infrared. - An infrared spectrogram shall be obtained using a procedure which has had prior approval of the Contracting Officer. ASTM E168 shall be used as a basis of the procedure.

4.5.2 Nitrogen. - The nitrogen shall be determined in accordance with Method 209.3 of Standard MIL-STD-286 or equivalent Method T 209.11.

4.5.3 Moisture. - The moisture shall be determined in accordance with Method 101.5 of Standard MIL-STD-286. See 6.6.



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4.5.4 Acidity or alkalinity. - The acidity or alkalinity shall be determined in accordance with Method 107.1 of Standard MIL-STD-286 using tolulene as the solvent.

4.5.5 Determination of stability. - The stability shall be determined as follows: Filter a portion of each DEGDN sample through two thicknesses "S and S." Number 604 filter paper or equivalent. Transfer a 2 mL portion of the filtered material by means of a pipette, to each of two test tubes which are 139.7mm long, 12.7mm internal diameter and 15.75mm outside diameter. (Care should be taken during the transfer not to leave droplets of the DEGDN on the sides of the test tubes.) Stopper each tube by means of a new, tightly fitting stopper through which passes a tightly fitted glass rod equipped with a platinum or glass holder for a strip of standard potassium iodide starch indicator paper. Using a forceps hang a strip of standard potassium iodide starch indicator paper on the platinum holder. The standard test papers shall be approximately 25mm long and 10mm wide. Moisten a horizontal section in the upper half of the standard test paper with a 50 percent solution (by volume) of pure glycerin in distilled water. This is conveniently accomplished by dipping a small diameter glass rod into the glycerin water solution, and as the rod is withdrawn, contact is made with the side of the container so as to minimize the volume of the solution adhering to the rod. Draw the rod across the paper strip so as to produce a level and distinct line of demarcation on the lower edge of the wet area. Prepare a blank tube by suspending a moistened strip of the standard test paper in a clean dry tube. Adjust the temperature of the heat tube bath to 82.2 $\pm$ 1°C and insert all three test tubes into the heating solution. The depth of immersion of the test tube into the heating solution should be approximately 50mm. The batch should be placed in such a position that the test tubes are viewed against a white background illuminated by bright diffused daylight. Note the time of insertion of the tubes into the bath. During the test the line of demarcation on the standard test paper should be 76mm above the level of the DEGDN in the test tube. The line of demarcation in the blank tube is regulated at an equivalent height in the tube. Consider the end point of the test to be the first appearance of any discoloration at the line of demarcation between the wet and dry portion of the test paper in the sample tube which is in excess of the discoloration



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observed at the same position on the test paper in the blank. Note the time for completion of the test to the nearest minute. Consider the minimum time for the two tubes to represent the heat value of the sample. After the test rinse all rods and tubes with acetone to remove DEGDN wash with warm soapy water, rinse thoroughly with tap water, then with distilled water and dry in a steam oven at approximately 80°C.

## 5. PACKAGING

5.1 Preservation, packing and packaging. - Diethylene Glycol Dinitrate is classed as a "Forbidden Explosive" in Code of Federal Regulations, Title 49, parts 100-199, for the Transportation of Explosives and Other Articles by Freight. For shipment by public highways see CFR, title 49, section 173.51 and Special Permit No. 5704 of the Department of Transportation (DOT) Hazardous Materials Regulations.

5.2 Marking. - The marking of interior packages and exterior shipping containers shall be in accordance with instructions contained in Special Permit No. 5704, Interstate Commerce Commission Regulations and in accordance with Standard MIL-STD-129.

## 6. NOTES

6.1 Intended use. - The DEGDN of this specification is intended for use as a propellant ingredient in the 120mm system.

6.2 Ordering data. - See MIL-A-48078.

6.3 Toxicity. - The DEGDN may cause dilation of blood vessels, headache, nausea, sinuses, and reduce blood pressure. Extreme exposures resulting from inhalation, skin contact or ingestion may result in central nervous system depression, coma, and respiratory paralysis. Drinking of any quantity of the liquids must be avoided. Severe headache shortly after exposure may diminish after development of tolerance. Ventilation should be used to reduce the inhalation hazard. Protective gloves and aprons are helpful in avoiding skin



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contact. Frequent washing in soap and water will minimize skin absorption. Alcohol or other solvents should not be used on the skin. The threshold limit value ceiling is 0.2 ppm.

6.4 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAR, Dover, New Jersey 07801.

6.5 Equivalent test methods. - Prior approval of the Contracting Officer is required for use of an equivalent test method. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method currently used versus the proposed method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Sampling for Infrared Spectrum. - At the start of any production run, an infrared spectrum shall be obtained for the first three batches, and every fifth batch thereafter. This procedure shall be mandatory especially when the facility is used for production of any material other than DEGDN.

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Project Number: 1315-A513



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